

TITANIC DATA CLEANING AND PREPROCESSING

Synent Technologies Data Science Internship

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1. Objective

The objective of this project is to clean and preprocess the Titanic dataset by handling missing values, removing duplicates, converting data types, and renaming columns.

1. Dataset Information

Dataset Name: Titanic Dataset

Source: Kaggle

Rows: 891

Columns: 12

1. Methodology

The following preprocessing steps were performed:

- Loaded dataset using Pandas
- Examined data structure and missing values
- Filled missing values in Age using median
- Filled missing values in Embarked using mode
- Removed Cabin column due to excessive missing values
- Checked and removed duplicate records
- Converted data types where necessary
- Renamed columns for better readability
- Results

The dataset was successfully cleaned.

Missing values were handled appropriately.

The cleaned dataset was exported as cleaned_titanic.csv.

1. Conclusion

Data preprocessing is an important step in the data science workflow. After cleaning, the Titanic dataset became suitable for further analysis and machine learning tasks.